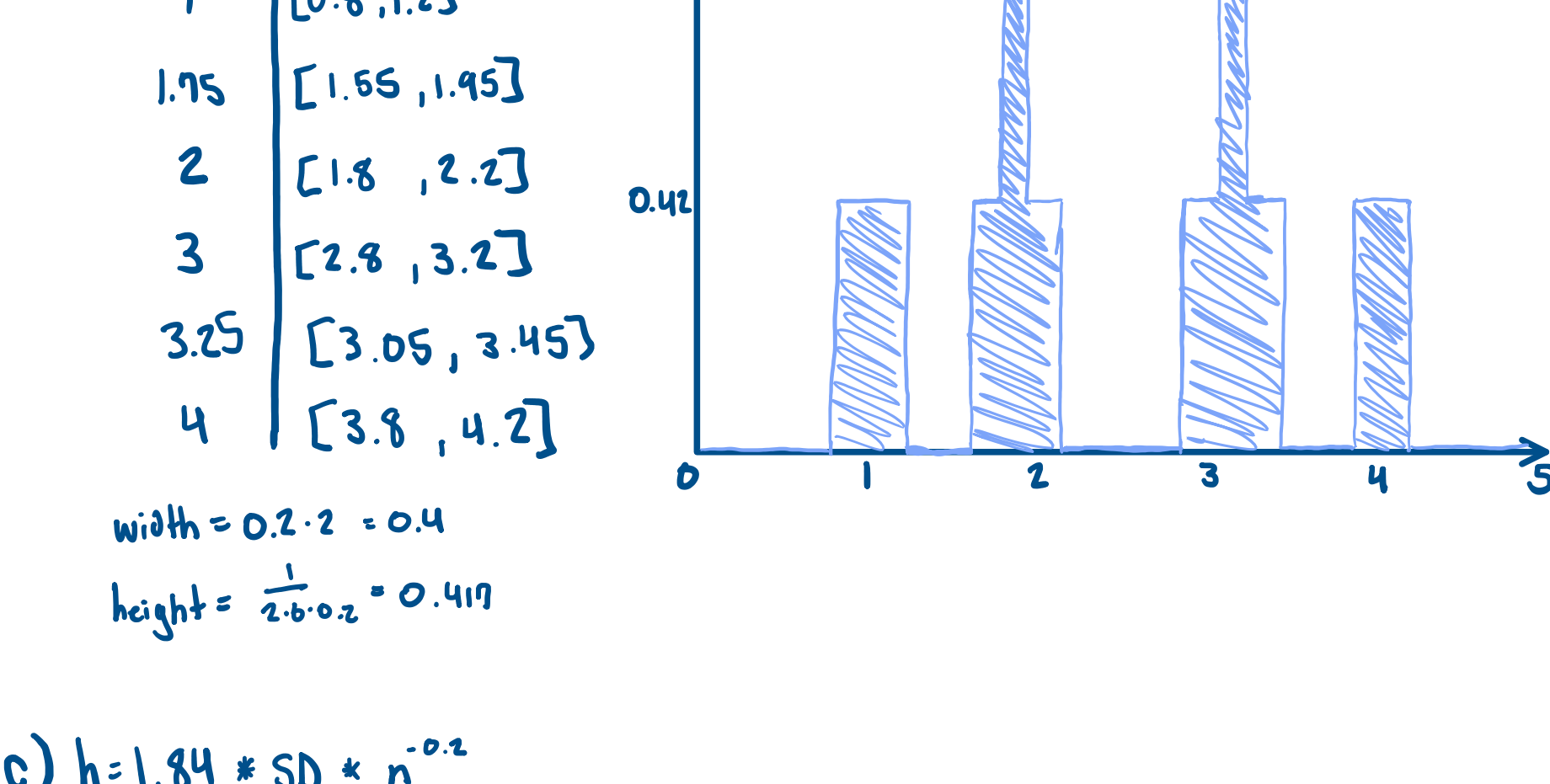
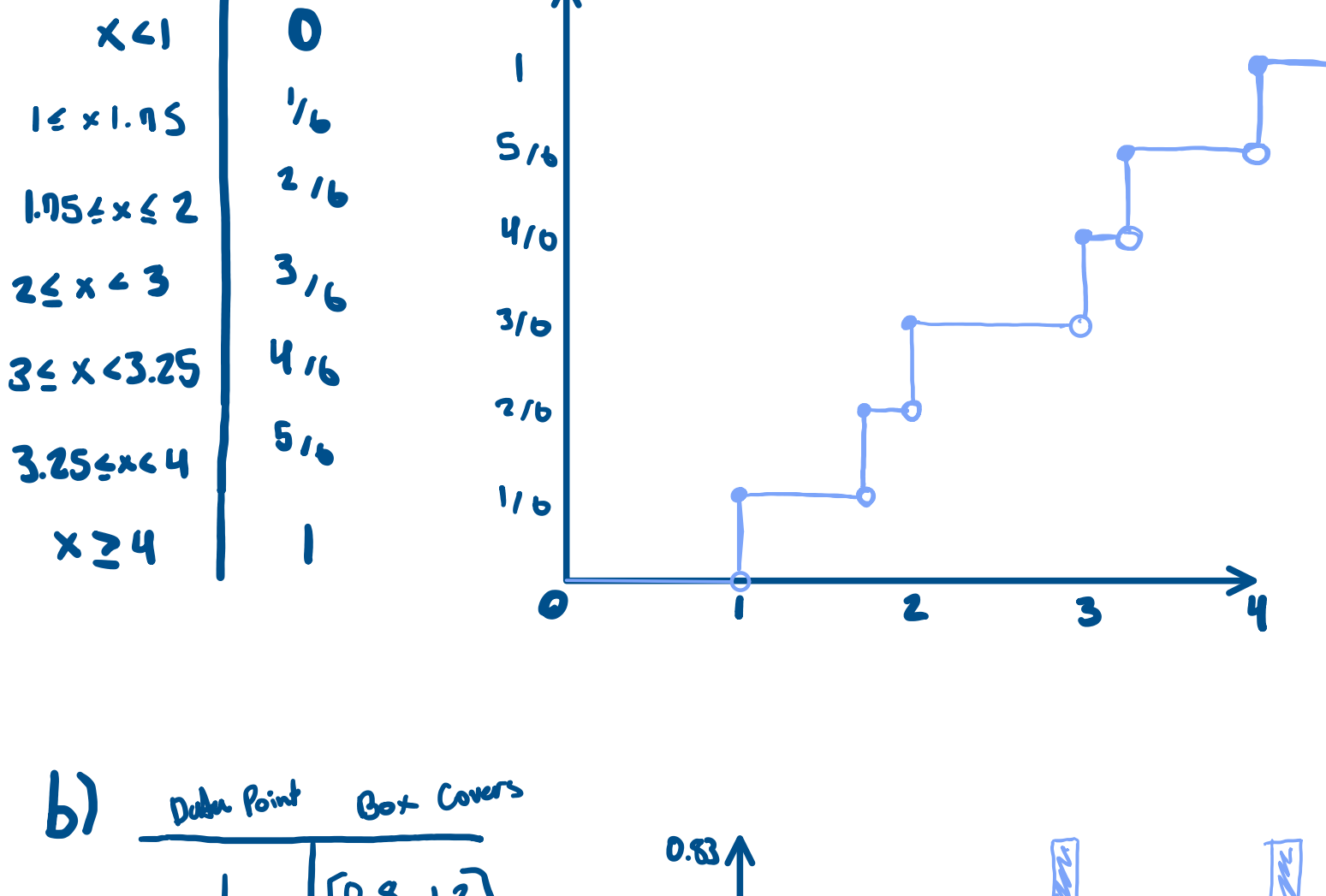


Problem 1

[1, 1.75, 2, 3, 3.25, 4]

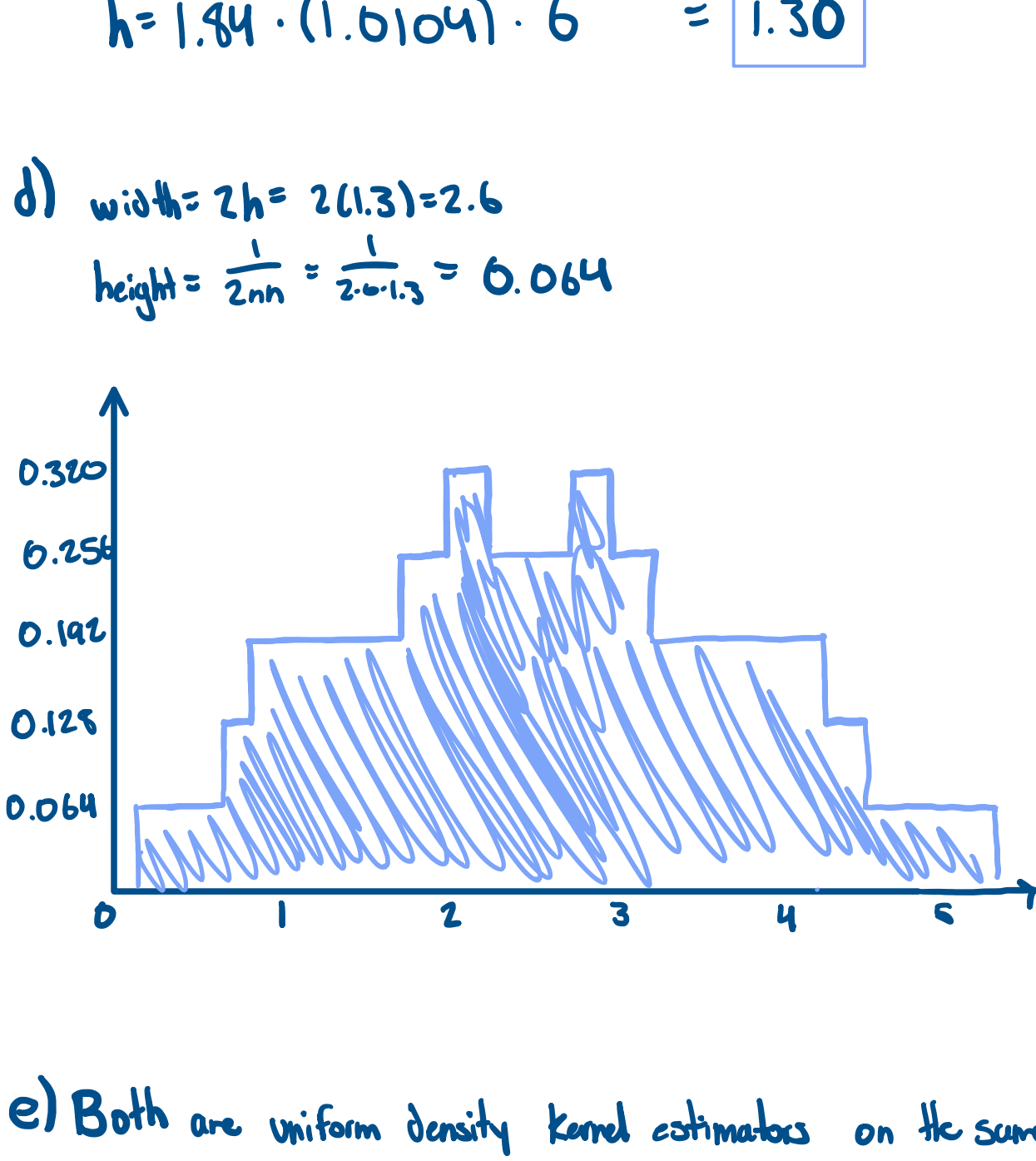


c) $h = 1.84 \cdot SD \cdot n^{-0.2}$
 $n = 6$
 $mean = \frac{1 + 1.75 + 2 + 3 + 3.25 + 4}{6} = \frac{15}{6} = 2.5$

SD	x	$x - mean$	$x - mean$ squared
	1	-1.5	2.25
	1.75	-0.75	0.5625
	2	-0.5	0.25
	3	0.5	0.25
	3.25	0.75	0.5625
	4	1.5	2.25

$\frac{2.25 + 0.5625 + 0.25 + 0.25 + 0.5625 + 2.25}{6} = \frac{6.125}{6} = 1.0208 = var(x)$
 $SD = \sqrt{1.0208} \approx 1.0104$

$h = 1.84 \cdot (1.0104) \cdot 6^{-0.2} = 1.30$



e) Both are uniform density kernel estimators on the same data. The difference between the two is their bandwidth h . Part b has a bandwidth of 0.2 so the box width is only 0.4. Only two pairs of points are close enough to overlap. Part d has $h = 1.3$ and the box width = 2.6 which is much wider than part b & wider than the full range of data. All b boxes overlap in this part. B underestimates relative to D.

Problem 2

[-1, -0.5, 0, 0.5, 1]

a) X_i | $10 - X_i$ | $10 - X_i \leq 0.6$?

-1	1	x
-0.5	0.5	✓
0	0	✓
0.5	0.5	✓
1	1	x

$\hat{f}_{0.6}(0) = \frac{3}{2 \cdot 5 \cdot 0.6} = \frac{3}{6} = 0.5$

b) $E[\hat{f}_h(0)]$?

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - x_i}{h}\right)$$

$$E[\hat{f}_h(x)] = E\left[\frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - x_i}{h}\right)\right]$$

$$= E\left[\frac{1}{h} K\left(\frac{x - x}{h}\right)\right]$$

Law of Unconscious Satisfaction

$$E[g(x)] = \int_{-\infty}^{\infty} g(t) f(t) dt \text{ where } f(t) \text{ is density of } x$$

$$E[\hat{f}_h(x)] = \int_{-\infty}^{\infty} \frac{1}{h} K\left(\frac{x-t}{h}\right) f(t) dt$$

Uniform kernel:

$$K(u) = \frac{1}{2} \text{ when } |u| \leq 1$$

$$\text{For } K\left(\frac{x-t}{h}\right) \neq 0,$$

$$\left|\frac{x-t}{h}\right| \leq 1$$

$$-1 \leq \frac{x-t}{h} \leq 1$$

$$h-x \leq -t \leq h-x$$

$$x-h \leq t \leq x+h$$

$$\int_{x-h}^{x+h} \frac{1}{h} K\left(\frac{x-t}{h}\right) f(t) dt \rightarrow \int_{x-h}^{x+h} \frac{1}{h} \cdot \frac{1}{2} f(t) dt \rightarrow \frac{1}{2h} \int_{x-h}^{x+h} f(t) dt$$

Standard Normal Density

$$\int_{x-h}^{x+h} f(t) dt = \Phi(x+h) - \Phi(x-h)$$

$$E[\hat{f}_h(x)] = \frac{\Phi(0.6) - \Phi(-0.6)}{2 \cdot 0.6}$$

$$= 0.376$$

c) $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$
 $f(0) = \frac{1}{\sqrt{2\pi}} e^{-0^2/2} = \frac{1}{\sqrt{2\pi}} e^0 = \frac{1}{\sqrt{2\pi}} = 0.3989$

d) No, B & C are not equal. $\hat{f}_h(x)$ is a biased estimator of $f(x)$ at $x=0$ with $h=0.6$. Since $E[\hat{f}_h(0)] \approx 0.376$ is below $f(0) \approx 0.399$, the estimator underestimates the true density here on average.

e) $E[\hat{f}_{0.1}(0)] = \frac{\Phi(0.1) - \Phi(-0.1)}{0.2} = \frac{0.5398 - 0.4602}{0.2} = 0.398$

- gap between $E[\hat{f}_h(0)]$ & $f(0)$ went from about -0.023 at $h=0.6$ down to about -0.001 at $h=0.1$. The gap decreased.

f) $\hat{f}_h$ is a consistent estimator at x if $\hat{f}_h(x) \rightarrow f(x)$ as the sample size grows, provided h is chosen correctly as n grows. h must shrink toward 0 to kill the bias since a smaller window means less averaging over the curve of the true density. nh must still grow toward infinity to kill the variance.

g) The main consideration is the tradeoff between bias & variance controlled by h . Small h means low bias but high variance and a jagged estimator. Large h means low variance but high bias and a smoother estimator.

Problem 3

$x = [5, 7, 7, 8, 10]$ $h = 1.5$

a) $x = 6$

X_i | $16 - X_i$ | within 1.5?

5 | 1 | ✓

7 | 1 | ✓

7 | 1 | ✓

8 | 2 | x

10 | 4 | x

$$f(6,0) = \frac{3}{16} = 0.2$$

$x = 6.5$

X_i | $16 - X_i$ | within 1.5?

5 | 1.5 | ✓

7 | 0.5 | ✓

7 | 0.5 | ✓

8 | 1.5 | ✓

10 | 3.5 | x

$$f(6.5) = \frac{4}{16} = 0.25$$

$x = 7$

X_i | $16 - X_i$ | within 1.5?

5 | 2 | x

7 | 0 | ✓

7 | 0 | ✓

8 | 1 | ✓

10 | 3 | x

$$f(7,0) = \frac{3}{16} = 0.2$$

$x = 8$

X_i | $16 - X_i$ | within 1.5?

5 | 3 | x

7 | 1 | ✓

7 | 1 | ✓

8 | 0 | ✓

10 | 2 | x

$$f(8,0) = \frac{3}{16} = 0.2$$

$x = 8.5$

X_i | $16 - X_i$ | within 1.5?

5 | 3.5 | x

7 | 1.5 | ✓

7 | 1.5 | ✓

8 | 0 | ✓

10 | 1.5 | ✓

$$f(8.5) = \frac{4}{16} = 0.25$$

b) $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$ $f_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$
 $x = 6$ $nh = 7.5$

x_i | $u = (6 - x_i)/1.5$ | $K(u)$

5 | 0.667 | 0.3195

7 | -0.667 | 0.3195

7 | -0.667 | 0.3195

8 | -1.33 | 0.1640

10 | -2.667 | 0.0114

$$1.1340 / 7.5 = 0.1512$$

$x = 6.5$

x_i | $u = 6.5 - x_i$ | $K(u)$

5 | 1 | 0.2420

7 | -0.33 | 0.3773

7 | -0.33 | 0.3773

8 | -1 | 0.2420

10 | -2.33 | 0.0262

$$1.2649 / 7.5 = 0.1686$$

$x = 7$

x_i | $u = 7 - x_i$ | $K(u)$

5 | 1.33 | 0.1640

7 | 0 | 0.3989

7 | 0 | 0.3989

8 | -0.667 | 0.3195

10 | -2 | 0.0540

$$1.3354 / 7.5 = 0.1781$$

$x = 8$

x_i | $u = 8 - x_i$ | $K(u)$

5 | 2 | 0.0540

7 | 0.667 | 0.3195

7 | 0.667 | 0.3195

8 | 0 | 0.3989

10 | -1.333 | 0.1640

$$1.2560 / 7.5 = 0.1675$$

$x = 8.5$

x_i | $u = 8.5 - x_i$ | $K(u)$

5 | 2.333 | 0.0262

7 | 1 | 0.242

7 | 1 | 0.242

8 | 0.333 | 0.3773

10 | -1 | 0.242

$$1.1295 / 7.5 = 0.1506$$

c) The biggest jump happens at $x=6.5$ and again at $x=8.5$. The uniform kernel has a flat top with hard edges, so a point is either fully in range of fully out. When x crosses that edge, the point contribution snaps on or off instantly which causes the jump. The Gaussian kernel is a smooth curve with no edges, so a point's weight just fades gradually as distance grows so no jumps occur.

d) The uniform kernel is better for simplicity & speed. It only requires counting points in a window. Gaussian kernel is better for accuracy & smoothness since it avoids jumps. It also gives a more realistic picture of the underlying density.

Problem 4

a)

b)

c) The main peak around 30-40 is reliable. The 15 resampled lines stack almost perfectly on top of each other there. The second bump around 55-65 shows noticeable variation.